INCH-POUND

MIL-A-85097/11B(AS)
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SUPERSEDING
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MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR ASSEMBLY, MK 113 MOD O

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 113 Mod 0 Rocket Motor Assembly described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification covers the requirements unique to the Rocket Motor Assembly Mk 113 Mod 0 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification to the extent specified herein.

SPECIFICATIONS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004 Propellant, NOSIH-BC-11

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL673AS200 Rocket Motor Assembly, Mk 113 Mod 0
DL673AS205 WORD Rocket Motor Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Attn: Code SR3, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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673AS200	Mk 113 Mod 0 WORD/Drogue Assembly
673AS205	WORD Motor Assembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

OP 2165	Navy Transportation Safety Handbook for Hazardous Materials

(Application for copies should be addressed to the Aviation Supply Office, Naval Publications and Forms Directorate, Attn: 1053, Building 26, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Rocket Motor Assembly Mk 113 Mod 0 shall conform to DL673AS200 and the Mk 113 Wind Oriented Rocket Deployment (WORD) Motor Assembly shall conform to DL673AS205.

3.2 Item description. The Rocket Motor Assembly Mk 113 Mod 0 specified herein is comprised of a WORD rocket motor mated to a drogue release mechanism.

3.3 Preproduction.

3.3.1 Propellant aging. When tested in accordance with MIL-A-85097, the WORD motor propellant shall meet the aging requirements specified in 511-174TD-004.

3.4 Preproduction/production.

3.4.1 WORD motor propellant. The propellant shall conform to 511-174TD-004. The contractor shall submit all test results in accordance with 511-174TD-004 and MIL-A-85097.

3.4.2 WORD motor ballistic performance. The ballistic performance shall conform to Table I.

3.4.3 Drogue release assembly performance. Upon application of 1,000 psig to either inlet port (see 4.1.3), the release shall actuate within 400 to 1,000 psig and shall in no way hinder deployment of the WORD rocket motor in any configuration/load arrangement.

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TABLE I. Ballistic performance requirements.

Characteristic	Value		Temperature (± 5•F)
	Min	Max	
Total impulse (lb-s)	47	55	-40, 160
Burn time (ms)	235	345	-40, 160
Ignition delay (ms)	-	10	-40, 160
Actuation force (lb)	15	25	-40, 160
Rise time (ms) (see 6.2.1)	-	30	-40, 160
Thrust (lb)	-	270	-40, 160

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 3 hours.
b. Temperature cycling: not less than 4 hours.

4.1.2 Radiographic inspection. Units shall be radiographed in two views 90• apart, as specified by Drawing 673AS200 (Mk 113 Mod O WORD/Drogue Assembly) or by Drawing 673AS205 (WORD Motor Assembly), as appropriate.

4.1.3 Shear pin testing. Shear pin testing shall be performed on the drogue release assemblies in accordance with the following:

a. The test units shall be conditioned at -40•F and 160•F for the time specified in 4.1.1a.

b. Test setup shall be as shown on Figure 1.

c. Actuation pressure shall be regulated to 1,000 to 1,025 and pressurization rate shall be regulated to 30,000 to 50,000 psig/sec.

d. Independent shear pin testing shall be conducted on each inlet port.

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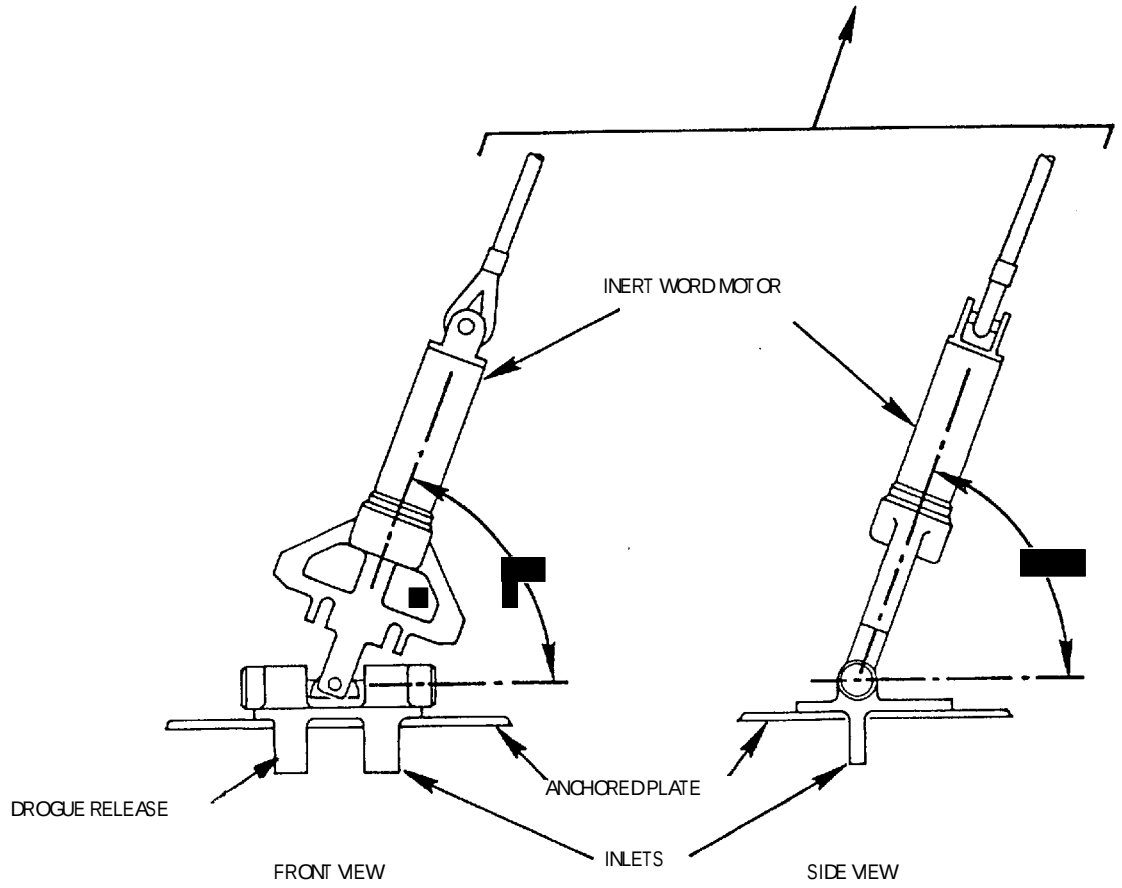

5. PACKAGING

5.1 Packaging. All packaging requirements shall be in accordance with MIL-A-85097 and the applicable contract.

FIGURE 1. Drogue release test configuration.

5.2 Marking. The shipping containers shall be marked "Explosive Power Device, Class B - Keep Fire Away" in accordance with OP 2165. See the applicable contract and the latest revision of MIL-A-85097 for additional marking requirements.

6. NOTES

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6.1 Intended use. The Mk 113 Mod 0 Rocket Motor Assembly provides means for drogue parachute release and subsequent personnel parachute deployment in an aircrew automated escape system.

6.2 Definition. For the purpose of this specification, the following definition applies:

6.2.1 Rise time. The interval from 10% to 75% of the maximum thrust during the initial rise on the thrust curve occurring just after ignition.

6.3 Changes from previous issue. Marginal notations are not used to denote changes from the previous issue because of the extensiveness of the changes.

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